

# STAT230A Lecture 7 Gauss-Markov Model (Parameter Estimation)

## 1 Gauss-Markov Model 的 Parameter Estimation

### Logic ▾

接下来我们研究对 Gauss-Markov model 的 parameters  $\beta$  和  $\sigma^2$  的 estimation

### Logic ▾

我们已经知道  $\hat{\beta} = (X^T X)^{-1} X^T \vec{Y}$  是  $\beta$  的 unbiased estimator, 我们希望进一步知道它是不是  $\beta$  最好的 unbiased estimator

### 1.1 Theorem: Gauss-Markov Theorem

对于 Gauss-Markov model,  $\hat{\beta}$  是 the **best linear unbiased estimator (BLUE)**, 即

对于任意满足以下条件的  $\tilde{\beta}$ :

- **Linear:**  $\tilde{\beta} = A\vec{Y}$  for some  $A \in \mathbb{R}^{p \times n}$  not depending on  $\vec{Y}$
- **Unbiased:**  $\mathbb{E}[\tilde{\beta}] = \beta$

有:

$$\text{Cov}(\hat{\beta}) \leq \text{Cov}(\tilde{\beta}) \quad (\text{matrix comparison})$$

#### Remark: Some Implications ▾

1. 若  $\text{Cov}(\hat{\beta}) \leq \text{Cov}(\tilde{\beta})$ , 则  $\text{Var}[\hat{\beta}_j] \leq \text{Var}[\tilde{\beta}_j]$ ,  $\forall j \in \{1, \dots, p\}$

#### Proof ▾

$$\begin{aligned} & \text{Cov}(\hat{\beta}) \leq \text{Cov}(\tilde{\beta}) \\ \Rightarrow & \text{Cov}(\tilde{\beta}) - \text{Cov}(\hat{\beta}) \succeq 0 \\ \Rightarrow & \vec{a}^T (\text{Cov}(\tilde{\beta}) - \text{Cov}(\hat{\beta})) \vec{a} \geq 0 \quad \forall \vec{a} \in \mathbb{R}^p \\ \Rightarrow & \vec{e}_j^T (\text{Cov}(\tilde{\beta}) - \text{Cov}(\hat{\beta})) \vec{e}_j \geq 0 \quad \forall j \in \{1, \dots, p\} \quad (\vec{e}_j = (0, \dots, 0, 1, 0, \dots, 0)^T) \\ \Rightarrow & \text{Var}[\hat{\beta}_j] \leq \text{Var}[\tilde{\beta}_j] \quad \forall j \in \{1, \dots, p\} \end{aligned}$$

2. Linear 是对于  $Y$  而言的 (不是对于  $X$ ),  $A$  可以是 non-linear function of  $X$ , 例如  $A = (X^T X)^{-1} X^T$
3.  $\tilde{\beta}$  必须是 any true  $\beta$  的 unbiased estimator, 因此无法选取  $A = 0$ , 一个潜在的  $\tilde{\beta}$  是 weighted linear square

#### Proof ▾

对于任意 unbiased estimator  $\tilde{\beta} = AY$ , 有

$$\begin{aligned} & \mathbb{E}[A\vec{Y}] = \beta \quad \forall \beta \\ \Rightarrow & A\mathbb{E}[\vec{Y}] = \beta \quad \forall \beta \\ \Rightarrow & AX\beta = \beta \quad \forall \beta \\ \Rightarrow & AX = I \end{aligned}$$

接下来证明  $\text{Cov}(\tilde{\beta}) - \text{Cov}(\hat{\beta}) \succeq 0$ , 注意到:

$$\begin{aligned}
\text{Cov}(\tilde{\beta}) - \text{Cov}(\hat{\beta}) &= \text{Cov}(\hat{\beta} + \tilde{\beta} - \hat{\beta}) - \text{Cov}(\hat{\beta}) \\
&= \text{Cov}(\hat{\beta}) + \text{Cov}(\tilde{\beta} - \hat{\beta}, \hat{\beta}) + \text{Cov}(\hat{\beta}, \tilde{\beta} - \hat{\beta}) + \text{Cov}(\tilde{\beta} - \hat{\beta}) - \text{Cov}(\hat{\beta}) \\
&= \text{Cov}(\tilde{\beta} - \hat{\beta}, \hat{\beta}) + \text{Cov}(\hat{\beta}, \tilde{\beta} - \hat{\beta}) + \text{Cov}(\tilde{\beta} - \hat{\beta})
\end{aligned}$$

由于对任意 random vector  $\vec{v}$ , 有  $\text{Cov}(\vec{v}, \vec{v}) \succeq 0$ , 因此:

$$\text{Cov}(\tilde{\beta} - \hat{\beta}) \succeq 0$$

而对于  $\text{Cov}(\hat{\beta}, \tilde{\beta} - \hat{\beta})$ , 记  $\hat{A}_{OLS} := (X^\top X)^{-1} X^\top$ , 有:

$$\begin{aligned}
\text{Cov}(\hat{\beta}, \tilde{\beta} - \hat{\beta}) &= \text{Cov}(\hat{A}_{OLS} Y, AY - \hat{A}_{OLS} Y) \\
&= \hat{A}_{OLS} \text{Cov}(Y)(A - \hat{A}_{OLS})^\top \\
&= \sigma^2 \hat{A}_{OLS} (A - \hat{A}_{OLS})^\top \\
&= \sigma^2 (\hat{A}_{OLS} A^\top - \hat{A}_{OLS} \hat{A}_{OLS}^\top) \\
&= \sigma^2 ((X^\top X)^{-1} X^\top A^\top - (X^\top X)^{-1} X^\top X (X^\top X)^{-1}) \\
&= \sigma^2 ((X^\top X)^{-1} (X^\top A^\top - I)) \\
&= 0 \quad (\text{由于 } AX = I)
\end{aligned}$$

类似地,

$$\text{Cov}(\tilde{\beta} - \hat{\beta}, \hat{\beta}) = 0$$

因此,

$$\begin{aligned}
\text{Cov}(\tilde{\beta}) - \text{Cov}(\hat{\beta}) &= \text{Cov}(\tilde{\beta} - \hat{\beta}) \succeq 0 \\
\Rightarrow \text{Cov}(\hat{\beta}) &\leq \text{Cov}(\tilde{\beta})
\end{aligned}$$

## Logic ▾

我们已经知道  $\hat{\beta}$  是  $\beta$  的 best linear unbiased estimator, 接下来我们研究  $\sigma^2$  的 unbiased estimator

## 1.2 $\sigma^2$ 的 Unbiased Estimator

对于 error variance  $\sigma^2$ , 其 unbiased estimator 为:

$$\hat{\sigma}^2 = \frac{\sum_{i=1}^n \hat{\varepsilon}_i^2}{n-p} = \frac{\vec{\hat{\varepsilon}}^\top \vec{\hat{\varepsilon}}}{n-p}$$

## Proof ▾

考虑 sum of  $\hat{\varepsilon}_i^2$ ,

$$\begin{aligned}
\mathbb{E} \left[ \sum_{i=1}^n \hat{\varepsilon}_i^2 \right] &= \sum_{i=1}^n \mathbb{E}[\hat{\varepsilon}_i^2] \\
&= \sum_{i=1}^n \text{Var}[\hat{\varepsilon}_i] \quad (\mathbb{E}[\hat{\varepsilon}_i] = 0) \\
&= \sum_{i=1}^n \sigma^2 (1 - h_{ii}) \\
&= \sigma^2 \left( n - \sum_{i=1}^n h_{ii} \right) \\
&= \sigma^2 (n - \text{Tr}(H)) \\
&= \sigma^2 (n - p) \quad (\text{若 } X^\top X \text{ 为 full rank})
\end{aligned}$$

因此,

$$\mathbb{E}[\hat{\sigma}^2] = \mathbb{E} \left[ \frac{\sum_{i=1}^n \hat{\varepsilon}_i^2}{n-p} \right] = \sigma^2$$